

MANOJKUMAR SANKAR

Software Engineer – Java Backend | Spring Boot | REST APIs | Microservices

Mayiladuthurai, India | +91 75488 94472 | manojmk2910@gmail.com | github.com/manojmkdev | linkedin.com/in/manojkumars2910

PROFESSIONAL SUMMARY

Results-driven Software Engineer with 2+ years of experience designing and delivering scalable, production-grade backend systems using Java 17, Spring Boot 3, MySQL, JWT Authentication, and RESTful API design. Skilled in role-based access control (RBAC), microservices architecture, Spring Security, and third-party integrations (Razorpay, Google Gemini AI, OAuth 2.0). Hands-on full-stack exposure with React. Strong foundation in OOP, data structures, algorithms, and SDLC. Seeking a challenging SDE / Java Backend Engineer role to build high-performance, real-world products.

TECHNICAL SKILLS

Languages: Java 17 (Primary), C++, C

Backend Frameworks: Spring Boot 3, Spring MVC, Spring Security, Spring Data JPA / Hibernate, RESTful API Design, Microservices

Authentication & Security: JWT (jjwt), BCrypt, Role-Based Access Control (RBAC), Google OAuth 2.0, OTP / SMTP Email Verification

AI & Integrations: Google Gemini AI (REST API), Razorpay Payment Gateway (order creation, webhook, verification)

Frontend: React 18, Redux Toolkit, HTML5, CSS3, JavaScript (ES6+), Bootstrap, Axios

Database: MySQL 8, JPA / Hibernate ORM, Relational Schema Design, Query Optimization

Tools & Platforms: GitHub, Postman, Swagger / OpenAPI, IntelliJ IDEA, VS Code, Maven, Vercel

Core Competencies: OOP, DSA, Problem Solving, System Design, Exception Handling, API Testing, SDLC, Agile

WORK EXPERIENCE

Software Engineer – Java Backend | Amypo Technologies, Coimbatore

March 2024 – Present

- Engineered and maintained production-grade full-stack web application modules using Java 17, Spring Boot 3, and MySQL with JWT-based stateless authentication across multi-role user hierarchies.
- Designed and implemented Role-Based Access Control (RBAC) dashboards with dedicated REST API workflows for Admin, Manager, and User roles, improving access governance and reducing unauthorized API calls.
- Architected and delivered backend modules aligned with SDE industry standards — structured service layers, DTO patterns, global exception handling, and input validation.
- Led development of backend LMS (Learning Management System) features, optimizing complex JPA queries and data flow to improve API response time and system throughput.
- Collaborated closely with front-end developers and product stakeholders in an Agile environment, contributing to sprint planning, code reviews, and technical documentation.
- Previously held the role of Skill Development Engineer at the same organization, leading internal technical training, onboarding processes, and developer tooling improvements.

Project Trainee – Java Full Stack Developer | Virtusa, Chennai

April 2023 – September 2023

- Contributed to building a mobile recharge platform (VirtuCharge) using Spring Boot with secure JWT authentication and role-based API authorization.
- Integrated Razorpay Payment Gateway end-to-end — including order creation, server-side transaction verification, and webhook / callback handling.
- Designed, tested, and debugged 20+ RESTful API endpoints using Postman for functional validation and regression testing.
- Gained production-level exposure to API security, error handling, and backend workflow design in a large enterprise codebase.

PROJECTS

QuizQuarry – AI-Driven Quiz & Proctoring Platform | *Java 17, Spring Boot 3, Spring Security, JWT, Google Gemini AI, MySQL, React 18, Redux Toolkit, Google OAuth 2.0*

- Built a full-stack AI-powered quiz platform where instructors generate MCQ question banks instantly using Google Gemini 2.5 Flash; questions include answer explanations and are fully editable before publishing.
- Implemented a proctored student assessment flow with per-question timers, tab-switch detection (Browser Visibility API), auto-submit on timeout, and configurable attempt limits enforced via Spring Security.
- Designed multi-role authentication (INSTRUCTOR / STUDENT) with JWT stateless security, Google OAuth 2.0 single sign-on, and OTP email verification (6-digit, 10-minute expiry via Gmail SMTP).
- Developed real-time analytics dashboards — instructor submission charts, student performance history, and a global leaderboard ranked by average score.
- Architected 12+ REST API endpoints (auth, quizzes, attempts, leaderboard) with full CRUD, rigorous input validation, and structured exception handling.

GitHub: github.com/manojmkdev/quizquarry

Admission Management CRM – College Admission Workflow System | *Java 17, Spring Boot 3, Spring Security, JWT, MySQL, HTML5, CSS3, Vanilla JS*

- Built an enterprise-grade College Admission CRM to manage end-to-end student admission workflows — institution setup, seat quota configuration, applicant lifecycle, document verification, and fee tracking.
- Implemented transactional seat allocation with @Transactional safety to prevent quota violations (HTTP 409 on overflow), ensuring zero double-booking across concurrent requests.
- Designed multi-role access (ADMIN, ADMISSION_OFFICER, MANAGEMENT) with Spring Security @PreAuthorize and JWT-based authentication; Management role restricted to read-only dashboards.
- Generated unique, immutable admission numbers encoding institution, year, program, quota type, and sequence — enforced at both application layer and database schema level (@Column unique + updatable=false).
- Developed a real-time dashboard REST API providing seat fill rates, pending document counts, and pending fee statistics across all programs and campuses.

GitHub: github.com/manojmkdev/Admission_Management_CRM_App

VirtuCharge – Mobile Recharge Platform | *Spring Boot, MySQL, Spring Security, Razorpay, JPA*

- Developed an end-to-end mobile recharge platform with secure authentication, plan management, and complete transaction lifecycle handling.
- Integrated Razorpay (Test Mode) for seamless payment flow — order creation, server-side verification, and callback handling.
- Designed RESTful APIs for CRUD operations on users, plans, and transactions with rigorous input validation and error handling.

Task Manager REST API | *Java, Spring Boot, Spring Security, JWT, MySQL*

- Built a secure task management API with JWT authentication for user registration, login, and personal task management with full CRUD operations.
- Implemented Spring Security filters, token validation, role-based access, and robust exception handling to protect all API endpoints.

GitHub: github.com/manojmkdev/Task_Manager_App-Spring-Boot-REST-API-JWT

CERTIFICATIONS

- Oracle Certified Associate – Java SE 8 Programmer | Oracle | February 2023

EDUCATION

B.Tech – Information Technology | Sri Krishna College of Technology, Coimbatore | April 2023 | CGPA: 8.67 / 10